# abstract-audit: TMLR figure and table sources

## Main text

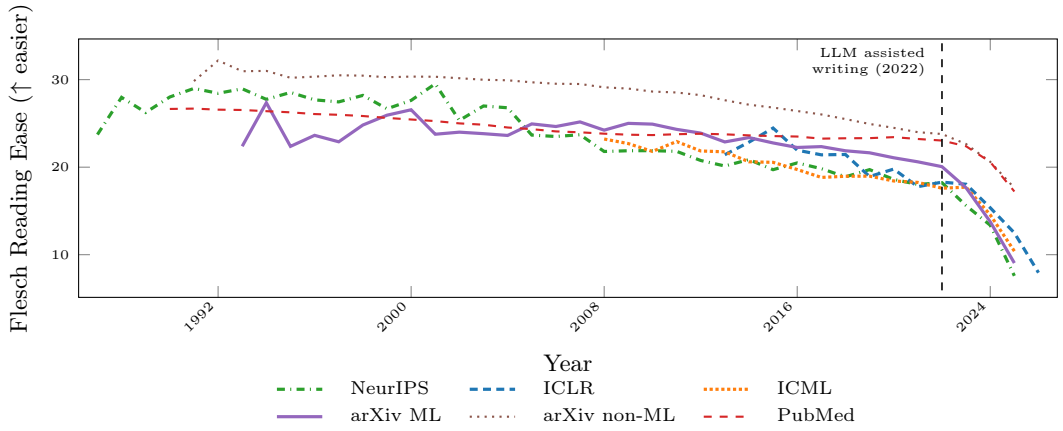



Figure 1: **Flesch Reading Ease across all six corpora (NeurIPS, ICLR, ICML, arXiv ML, arXiv non-ML, and PubMed), 1987–2026.** Higher values indicate easier reading (↑). The three conferences and arXiv ML all decline, and by 2025 each sits below both the arXiv non-ML and the PubMed baseline.

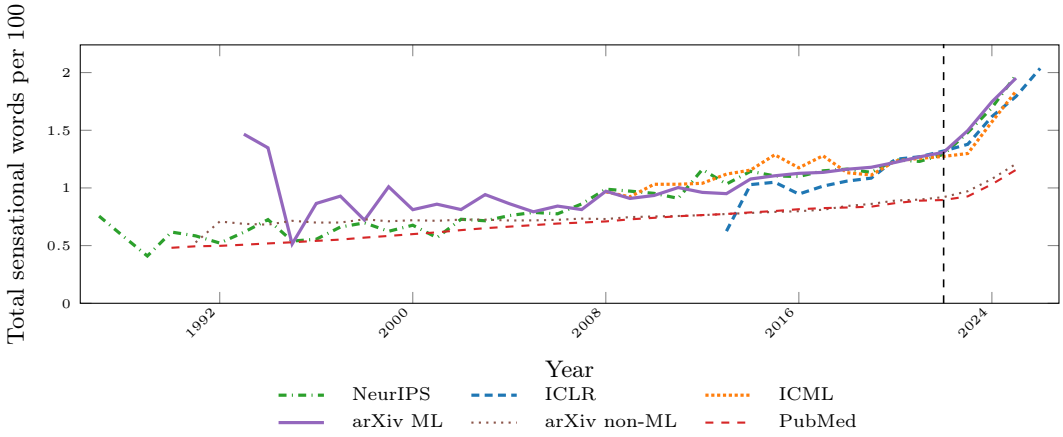



Figure 2: **Total sensational language per 100 abstract words, 1987–2026.** The rate rises across all three conferences and arXiv ML. The arXiv non-ML baseline rises far more slowly, and the divergence between ML and non-ML venues accelerates after 2022. The breakdown by category appears in Appendix **??**.

| Family | What it measures | Section | Direction over time |
|---|---|---|---|
| Classical readability, 15 formulas | Surface difficulty of the prose, from sentence length and word length | ?? | Harder to read at every venue |
| Hohmann writing style suite, extended to 12 metrics | Syntactic structure, including parse depth, passive rate and noun phrase density | ?? | Largely parallel to the non-ML control |
| Sensational language rate, part of the style suite | Rhetorical stance, counted as promotional words per 100 words | ?? | Rising at every venue |
| Acronym density and reuse | Vocabulary a reader must already hold, and how often each term recurs | ?? | Denser, with most terms rarely reused |
| LLM as judge, 6 models | What an automated reviewer reports about readability | ?? | Rising after 2022 at NeurIPS and ICML, against the classical scores |
| Citation counts | What the field rewarded, within venue-year cohorts | ?? | More readable papers cited slightly more |

Table 1: The five metric families, what each one measures, and the direction it moves at NeurIPS, ICLR and ICML.

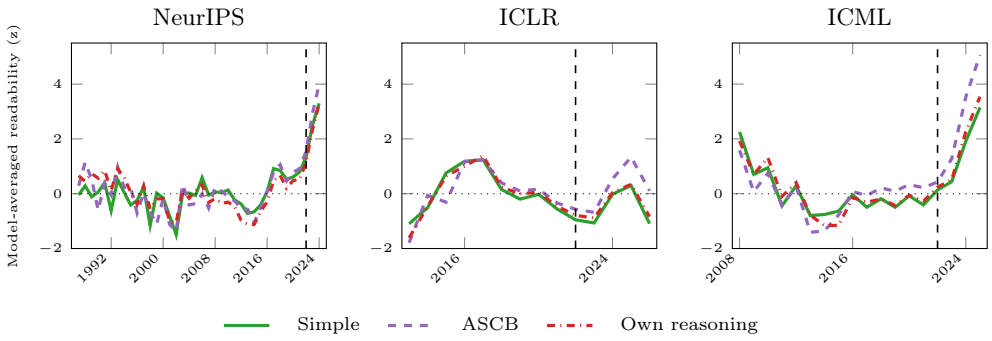



Figure 3: **LLM as judge readability across the three conferences.** Each curve is the mean z-score across all six open-weight models for one prompt variant, standardized to each model's own baseline through 2022 within that venue. NeurIPS and ICML show the same pattern. Scores rise sharply after 2022, by roughly three to five standard deviations, directly contradicting the classical readability decline in Figure 1. ICLR shows no comparable rise, and its post-2022 movement does not settle into a consistent trend across the three prompts, so we draw no conclusion from that series and leave it to further investigation.

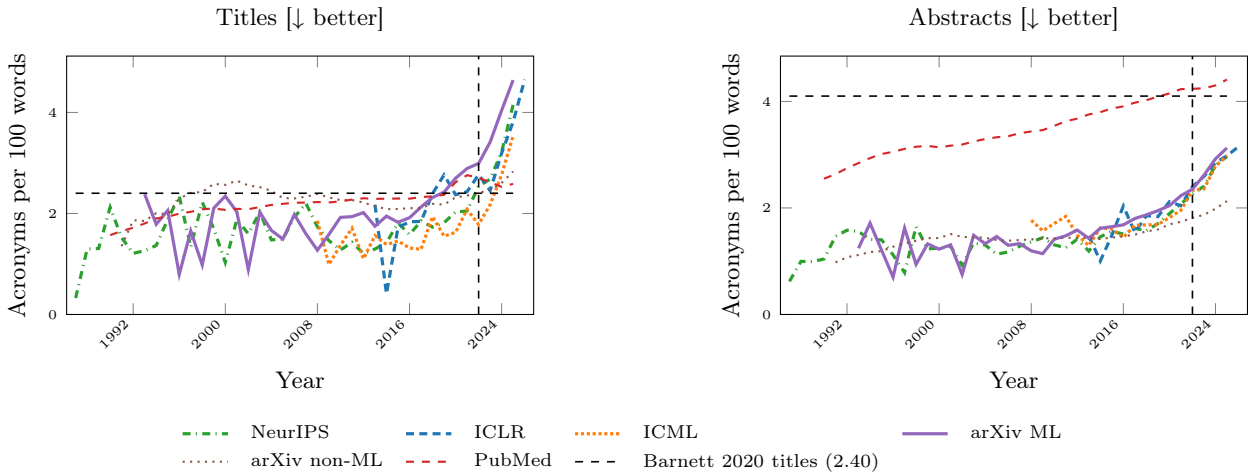



Figure 4: **Acronym density in titles (left) and abstracts (right), 1987 to 2026. ?** ] report acronym density for titles and for abstracts separately, and their 2019 biomedical values differ between the two. The dashed black horizontal line marks the applicable one in each panel, 2.40 per 100 words for titles and 4.10 for abstracts. ML title density has risen above the biomedical title value, while ML abstract density remains below the biomedical abstract value.

Table topics.tex not rendered: the citation outputs were not built (the citation counts are not distributed; see README, Citations)

# Appendix

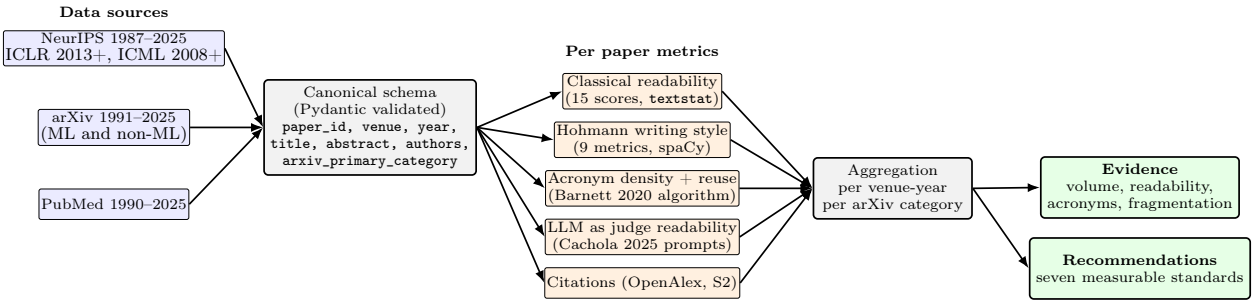



Figure 5: **End to end workflow.** Three data sources feed a canonical Pydantic-validated schema. Five per-paper metric families are computed on abstracts, and acronym density also on titles, then aggregated per venue-year and per arXiv primary category. The outputs feed the empirical evidence (Sections **??** through **??**) and the recommendations (Section **??**). Pipeline details appear in Appendix **??**.

| Source | Coverage | Papers |
|---|---|---|
| arXiv (all categories) | 1991–2025 | 2,823,623 |
| NeurIPS | 1987–2025 | 30,595 |
| ICLR | 2013–2026 | 16,807 |
| ICML | 2008–2025 | 15,226 |
| PubMed | 1990–2025 | 24,306,644 |

Table 2: Per source paper counts and coverage at the 2026-04 snapshot. The arXiv 2025 row covers approximately three quarters of the calendar year. The NLM baseline additionally carries 156,564 ahead-of-print PubMed records dated 2026 and 2027. An ahead-of-print date is not a publication year, so these are excluded from the PubMed count above and from every year-indexed analysis and figure, and from the pooled acronym reuse distribution of Table 8.

| Element | Before | After |
|---|---|---|
| Inline mathematics | `coresets ($\texttt{FWC}$)` | `coresets ( x )` |
| Display mathematics | `regret is $O(\sqrt{T})$` | `regret is x` |
| Mathematics spliced inside a word | `$\underline{Co}$llabo-$\underline{ra}$tive` | `Collaborative` |
| Text-mode commands | `the \emph{\textbf{same}} setting` | `the same setting` |
| URLs | `available at https://-github.com/ex/repo` | `available at` |
| Identifiers | `see arXiv:2401.00001` | `see` |
| Ampersands | `speed & accuracy` | `speed and accuracy` |
| Non-ASCII characters | `the café corpus` | `the caf corpus` |
| Section headings | `OBJECTIVE: We study scaling.` | `We study scaling.` |

Table 3: Text normalisation rules, with the effect of each on a worked example. Applied for the sensitivity analysis in Table 4. The reported metrics are computed on the abstract as published.

| Normalisation | Venue | First year | Last year | Change |
|---|---|---|---|---|
| None | NeurIPS 1987–2025 | 23.73 | 7.59 | −16.14 |
| LaTeX only | NeurIPS 1987–2025 | 23.74 | 7.92 | −15.82 |
| LaTeX and URLs | NeurIPS 1987–2025 | 23.74 | 8.44 | −15.31 |
| None | ICLR 2013–2026 | 21.42 | 7.94 | −13.48 |
| LaTeX and URLs | ICLR 2013–2026 | 21.52 | 8.74 | −12.78 |
| None | ICML 2008–2025 | 23.22 | 10.41 | −12.81 |
| LaTeX and URLs | ICML 2008–2025 | 23.22 | 11.23 | −11.99 |

Table 4: Flesch Reading Ease at each venue's first and last year under three text normalisations. Markup accounts for 5 to 6 percent of each measured decline, and URLs for more of it than LaTeX.

**Simple Prompt**

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. Assume the reader is an adult. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

**ASCB Guidelines Prompt**

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. Characteristics of a highly readable text include: \n
– Know your audience, and focus and organize your information for that particular audience. \n
– Focus on the big picture. What larger problem is your work a part of? What major ideas or issues does your work address? How will your work help global understanding of some issue? \n
– Avoid jargon. If you must use a technical term, make sure to explain it, but simplify the language. \n
– Try to use metaphors or analogies to everyday experiences that people can relate to. \n
– Underscore the importance of public support for exploratory research and scientific information, and the role of this information in providing the context for effective policy making. \n \n
Assume the reader is an adult. Do not use Flesch-Kincaid or other readability formulas. Use your own judgment to rate the text. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

**Own Reasoning Prompt**

On a scale of 1 to 5, what is the reading ease of the following text? 1 indicates the text requires expert background knowledge and 5 indicates the text is readable to the general population. \n
Assume the reader is an adult. Do not use Flesch-Kincaid or other readability formulas. Use your own judgment to rate the text. \n \n
Format the output as follows: \n
Score: <score> \n
Text: {SUMMARY}

Table 5: The three prompt templates we apply, reproduced verbatim from **?** ] (Table 10) except that the Reason: <reasoning> output line is omitted. {SUMMARY} is replaced by the abstract text at inference.

| ID | Leading terms | Papers | Span | ΔFlesch | ΔHype | ΔAcronyms |
|---|---|---|---|---|---|---|
| T3 | task, tasks, domain, data, performance, knowledge | 4,957 | 2015–2025 | −16.7 | +0.56 | +1.90 |
| T12 | graph, graphs, gnns, graph neural, node, networks | 2,250 | 2018–2025 | −13.2 | +0.76 | +2.35 |
| T2 | reasoning, llms, language, language models | 2,572 | 2021–2025 | −12.2 | +0.49 | +1.48 |
| T8 | language, llms, language models, large language | 4,042 | 2020–2025 | −11.3 | +0.59 | +1.24 |
| T5 | inference, latent, bayesian, data, variational | 3,685 | 2015–2025 | −11.1 | +0.32 | +0.48 |
| T11 | neural, neurons, network, brain, networks, visual | 891 | 2016–2025 | −11.1 | +0.41 | +0.80 |
| T4 | 3d, video, object, scene, view, image, objects | 2,442 | 2018–2025 | −11.0 | +1.20 | +1.37 |
| T13 | networks, neural, neural networks, network, deep | 5,065 | 2015–2025 | −10.5 | +0.38 | +0.78 |
| T1 | diffusion, image, diffusion models, generation | 2,958 | 2018–2025 | −9.6 | +0.90 | +0.30 |
| T15 | regret, algorithm, online, bandit, algorithms | 2,336 | 2015–2025 | −8.8 | +0.29 | +0.67 |
| T7 | data, machine, machine learning, training | 5,255 | 2015–2025 | −8.4 | +0.19 | +1.20 |
| T10 | supervised, self, supervised learning | 2,045 | 2016–2025 | −7.8 | +0.78 | +0.91 |
| T6 | reinforcement, reinforcement learning, policy, rl | 4,445 | 2016–2025 | −7.7 | +0.49 | +1.73 |
| T9 | gradient, optimization, convergence, convex | 2,760 | 2015–2025 | −5.7 | +0.30 | +0.52 |
| T14 | algorithm, problem, data, algorithms, linear | 4,457 | 2015–2025 | −4.6 | +0.13 | +0.47 |

Table 6: Change in three of the five metric families within each topic, measured from the first to the last year in which that topic has at least 30 papers within the 2015 to 2025 window. Several topics have fewer than 30 papers at the start of that window. Flesch Reading Ease is in points, sensational language and acronym density in occurrences per 100 words. Paper counts are for the window, so they are lower than the full-corpus counts in Table **??**, whose identifiers these match.

Table topic_controls.tex not rendered: the citation outputs were not built (the citation counts are not distributed; see README, Citations)

| Weighting | Within topic | Between topic | Within share (%) |
|---|---|---|---|
| Topic mix held at 2015 | −9.1 | −1.8 | 83 |
| Mix averaged over both years | −13.3 | +2.4 | 122 |
| Topic mix held at 2025 | −17.6 | +6.7 | 161 |
| Gap to papers of the same year | −11.0 | +0.1 | 101 |

Table 7: The fall of 10.9 points in mean Flesch Reading Ease across the three conferences from 2015 to 2025, split into a part within topics (topics becoming less readable) and a part between topics (the field moving between topics). Each row sums to the total. The first three rows value each topic at its own mean score and differ only in which year's topic mix weights the within-topic change. The last row values each topic by its gap to papers of the same year, estimated over every year of the window, which avoids topic means that rest on a few papers in 2015. A positive between-topic entry means the change in topic mix raised readability.

| Venue | Unique acronyms | 1 | 2–9 | 10–99 | 100–999 | 1000+ |
|---|---|---|---|---|---|---|
| NeurIPS | 15,693 | **31.2%** | 58.3% | 9.5% | 1.0% | 0.0% |
| ICLR | 11,097 | **27.0%** | 63.2% | 8.8% | 0.9% | 0.0% |
| ICML | 8,218 | **26.9%** | 62.8% | 9.6% | 0.7% | 0.0% |
| arXiv ML | 115,229 | **28.9%** | 59.1% | 10.3% | 1.5% | 0.2% |
| arXiv non-ML | 181,570 | **32.0%** | 50.7% | 13.7% | 3.1% | 0.6% |
| PubMed | 1,679,954 | **31.1%** | 48.4% | 16.4% | 3.2% | 0.9% |
| Barnett and Doubleday (2020) | 1,112,345 | 30.0% | 49.0% | n/a | n/a | n/a |

Table 8: Acronym frequency distribution by venue (abstract acronyms). Each cell shows the percentage of unique acronyms appearing in that frequency range. The external baseline is Barnett and Doubleday (2020), PubMed titles and abstracts, 1950–2019. Single-use rates are comparable across venues. ML venues carry a larger 2–9 use bucket. Roughly 88 to 90 percent of ML acronyms appear fewer than ten times against 80 percent in PubMed. A value of n/a means that the source does not report the frequency range.

| Acronyms | Share of distinct acronyms (%) | Share of occurrences (%) |
|---|---|---|
| Top 10 | 0.04 | 15.8 |
| Top 20 | 0.09 | 21.4 |
| Top 50 | 0.22 | 30.7 |
| Top 100 | 0.45 | 38.4 |
| Top 500 | 2.24 | 56.9 |

Table 9: Concentration of abstract acronym occurrences at the three conferences between 2015 and 2025. The corpus holds 22,332 distinct acronyms over 229,789 occurrences. A few hundred reused terms account for most of what a reader meets, while the 25.6 percent of acronyms that occur once carry only 2.5 percent of occurrences.

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Table citation_llm_top_bottom_combined.tex not rendered: the citation outputs were not built (the citation counts are not distributed; see README, Citations)

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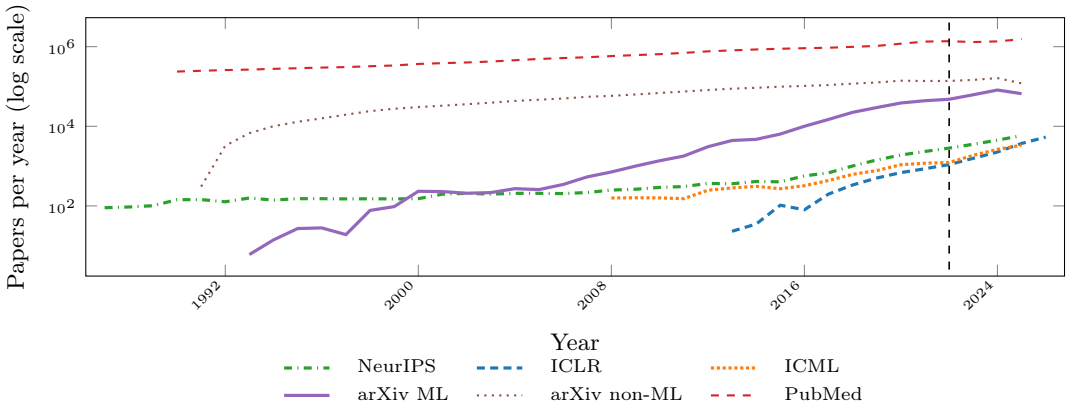



Figure 6: **Papers per year, 1987 to 2026.** arXiv ML submissions grew by roughly four orders of magnitude between 1993 and 2025. NeurIPS grew by a factor of roughly 65 between 1987 and 2025, from fewer than 100 accepted papers to approximately 5,800. PubMed and arXiv non-ML show sustained but slower growth over the same period. The volume increase is the structural backdrop against which the readability and acronym trends in Sections **??** and **??** must be interpreted.

| Metric | Primary sources |
|---|---|
| Sentence length | **?** **?** ] |
| Sign posting | **?** **?** ] |
| Hedging | **?** **?** **?** ] |
| Noun chunks | **?** **?** **?** ] |
| Active narration | **?** **?** **?** ] |
| Verbs | **?** **?** ] |
| Nouns | **?** ] |
| Numbers | **?** ] |
| Sensational language | **?** ] |

Table 10: Writing style metrics audited per paper, with the original sources cited by **?** ].

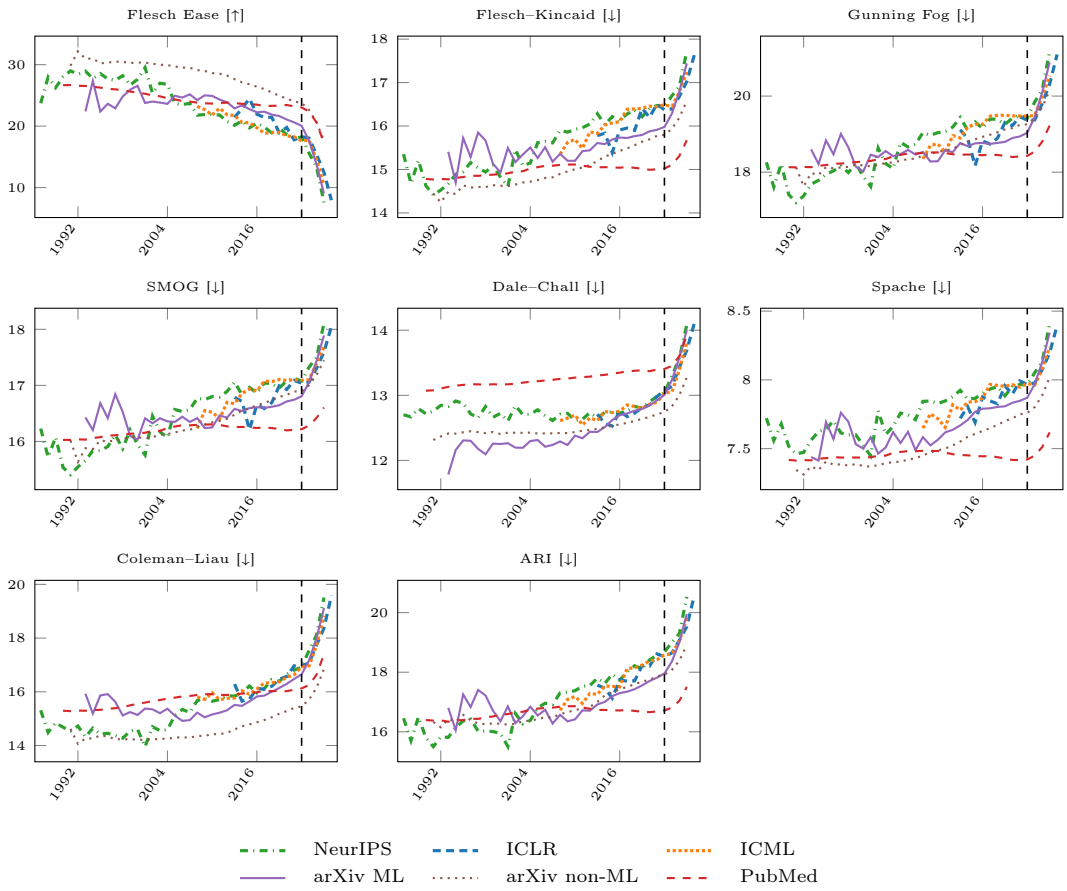



Figure 7: **Classical readability metrics, part A: Flesch family and grade-level formulas, 1987–2026.** Arrows indicate the direction of easier reading (↑ = higher is better, ↓ = lower is better). Every grade-level metric rises on NeurIPS and arXiv ML after 2015 and accelerates after 2020. Flesch Reading Ease drops correspondingly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part B with the remaining seven metrics appears in Figure 8.

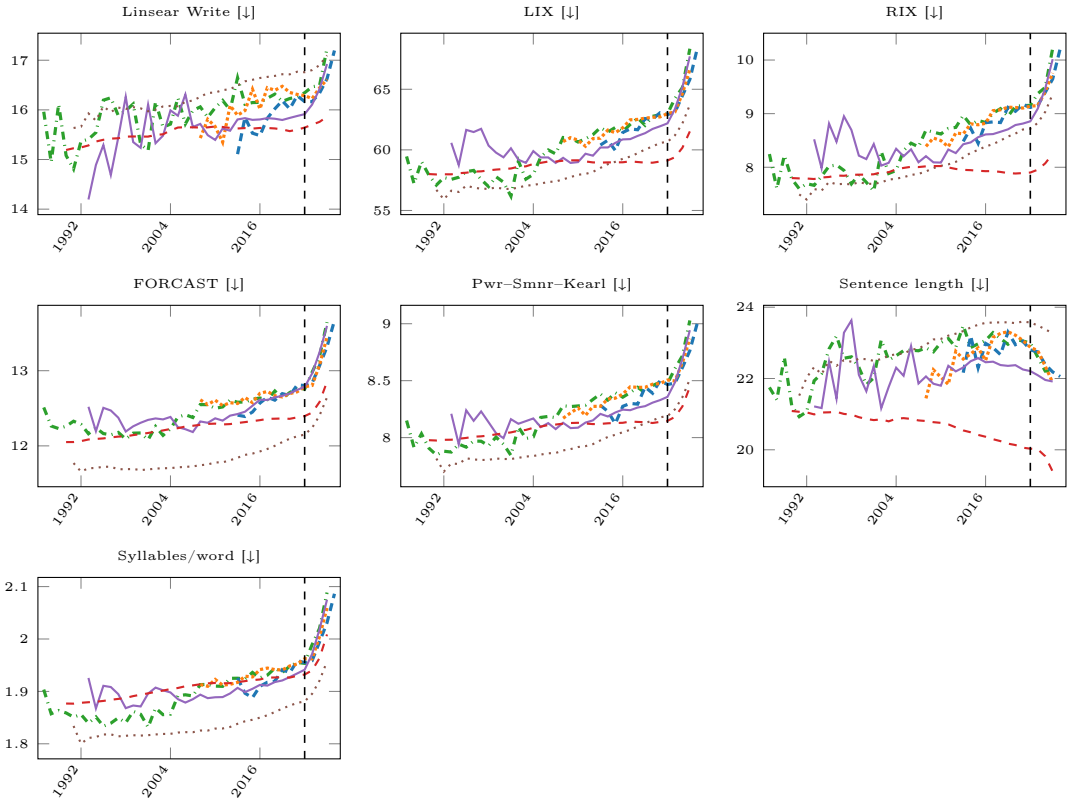



Figure 8: **Classical readability metrics, part B: Linsear Write, LIX, RIX, FORCAST, Powers–Sumner–Kearl, sentence length, and syllables per word, 1987–2026.** Arrows indicate the direction of easier reading (↓ = lower is better). Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants. Part A with the Flesch family and the first eight grade-level formulas appears in Figure 7.

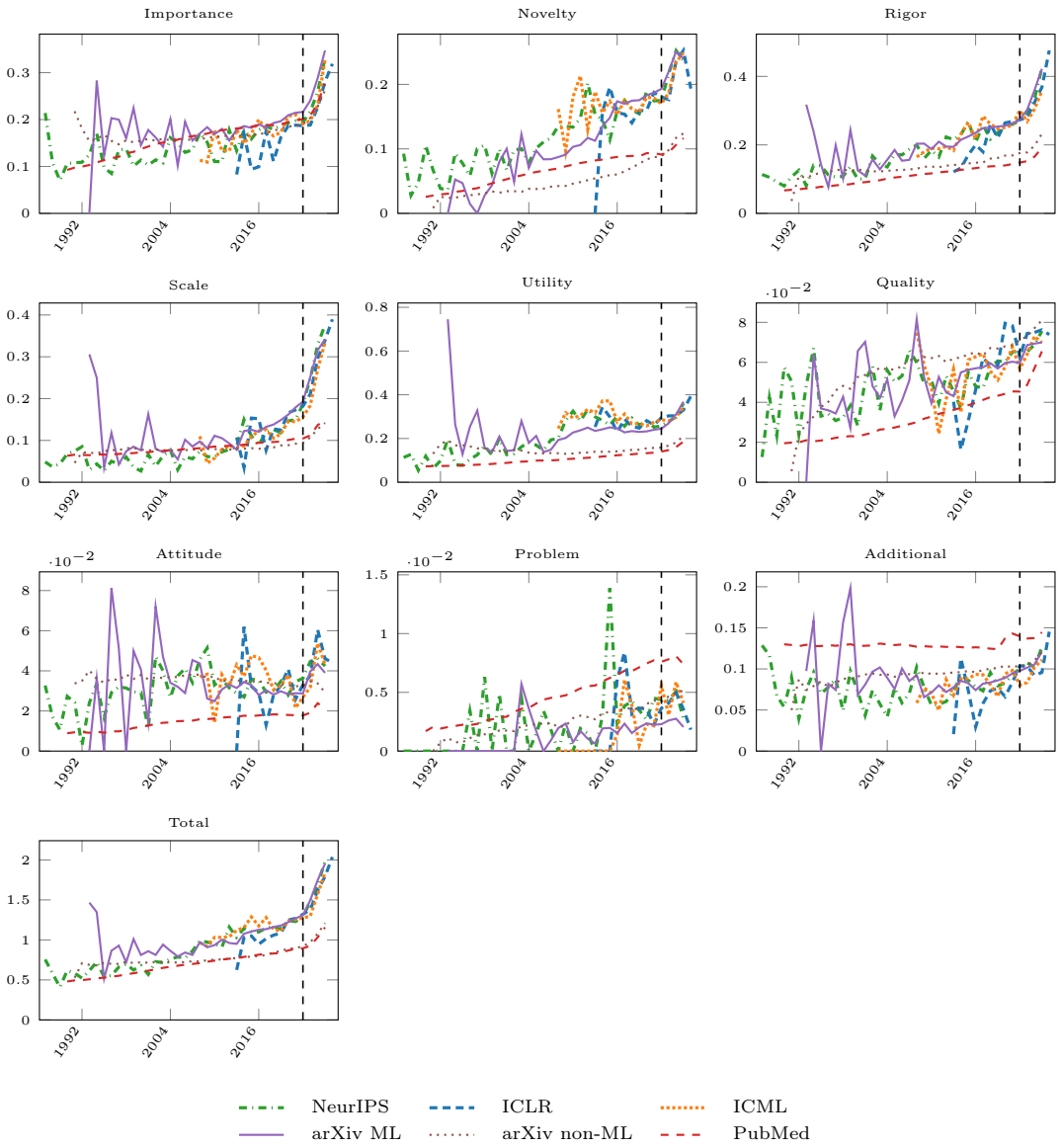



Figure 9: **All nine sensational language categories and their total per 100 abstract words, 1987–2026.** Novelty and scale rise sharply after 2015, reaching roughly 2.1 times and 3.4 times their 2015 levels on NeurIPS by 2025 (1.6 and 2.7 times on arXiv ML). The total rate and rigor rise more modestly. Vertical dashed lines mark the late-2022 availability of instruction-tuned writing assistants.

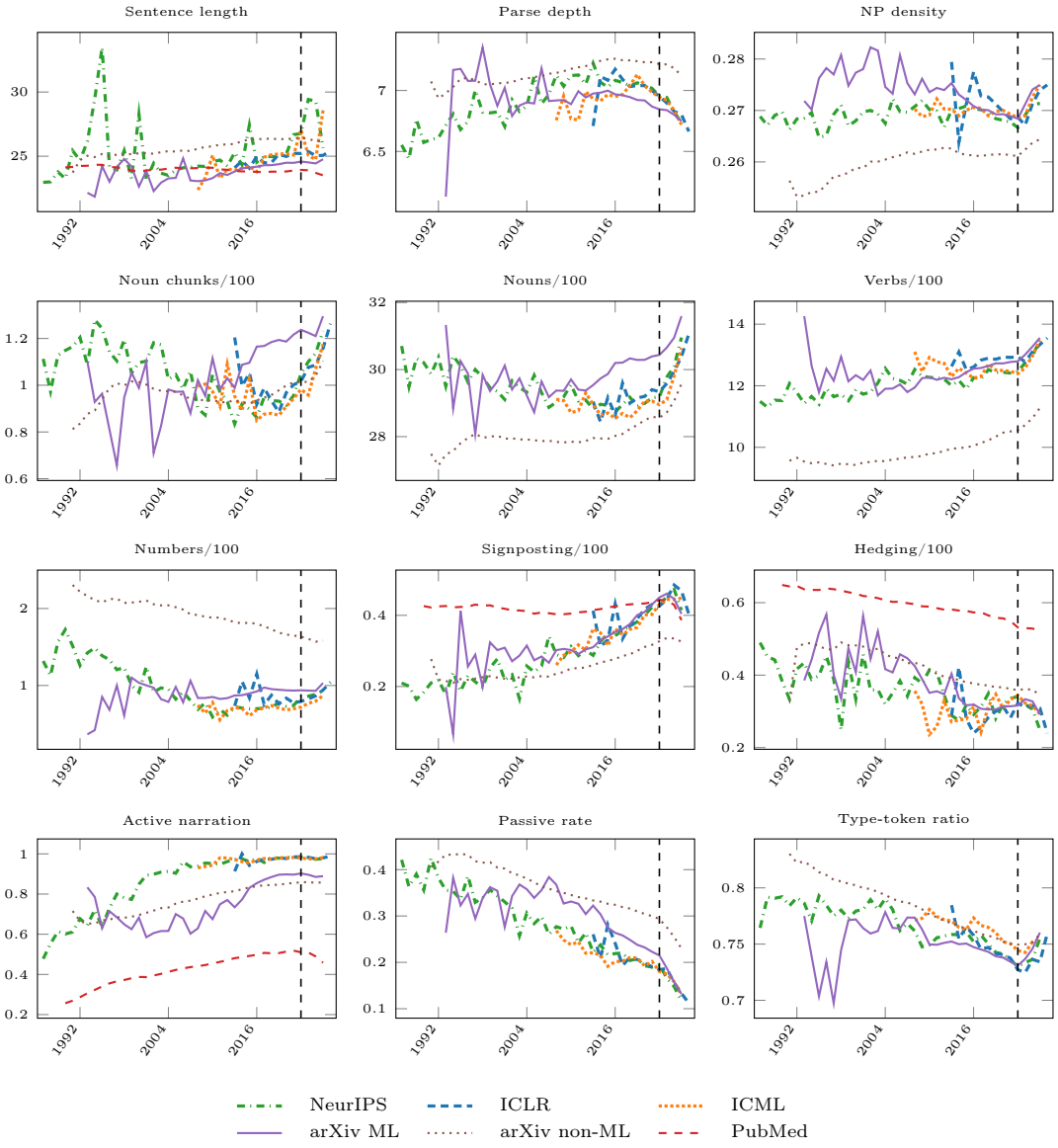



Figure 10: **Hohmann writing style metrics across all six corpora (NeurIPS, ICLR, ICML, arXiv ML, arXiv non-ML, and PubMed), 1987–2026.** PubMed data are available only for the four metrics computed by lightweight regex and word-level rules (sentence length, signposting, hedging, and active narration). The remaining metrics depend on full spaCy dependency parsing and were not run on the 24.3 million paper PubMed corpus because of compute cost. The corresponding panels omit PubMed. Across the suite, ML and non-ML venues are largely parallel.

| Metric | A | Pct |
|---|---|---|
| **Classical readability (15)** | | |
| Flesch Reading Ease | 61.50 | 99.9% |
| Flesch–Kincaid Grade | 7.80 | 0.0% |
| Gunning Fog | 9.31 | 0.0% |
| SMOG | 9.76 | 0.0% |
| Dale–Chall | 12.73 | 31.3% |
| Spache | 5.09 | 0.0% |
| Coleman–Liau | 10.47 | 0.5% |
| ARI | 8.97 | 0.1% |
| Linsear Write | 7.43 | 0.2% |
| LIX | 38.61 | 0.1% |
| RIX | 3.27 | 0.1% |
| FORCAST | 10.61 | 0.6% |
| Powers–Sumner–Kearl | 5.91 | 0.1% |
| Words per sentence | 12.54 | 0.3% |
| Syllables per word | 1.57 | 0.3% |
| **Acronyms** | | |
| Acronym density /100 | 5.12 | 89.0% |
| Novel acronyms (title) | 0 | n/a |
| Novel acronyms (abstract) | 0 | n/a |

| Metric | A | Pct |
|---|---|---|
| **Writing style suite (12)** | | |
| Sentence length (Hohmann) | 22.93 | 35.7% |
| Parse depth | 5.47 | 6.7% |
| NP density | 0.24 | 12.2% |
| Noun chunks /100 | 0.87 | 49.2% |
| Nouns /100 | 24.13 | 7.2% |
| Verbs /100 | 9.01 | 6.3% |
| Numbers /100 | 13.95 | 100.0% |
| Sign posting /100 | 0.00 | 24.3% |
| Hedging /100 | 0.58 | 77.1% |
| Active narration | 1.00 | 52.2% |
| Passive rate | 0.00 | 14.4% |
| Type–token ratio | 0.66 | 13.3% |
| **Sensational language (10)** | | |
| Importance | 0.00 | 34.4% |
| Novelty | 0.00 | 35.9% |
| Rigor | 0.00 | 33.2% |
| Scale | 0.00 | 35.3% |
| Utility | 0.00 | 32.3% |
| Quality | 0.00 | 44.7% |
| Attitude | 0.00 | 46.7% |
| Problem | 0.00 | 49.7% |
| Additional emphasis | 0.00 | 42.3% |
| Total | 0.00 | 8.6% |

Table 11: Metric self-audit of this manuscript. Numbers are produced by running the same per-paper metric scripts used throughout the paper on this paper's own title and abstract. Percentile refers to the NeurIPS 1987–2025 abstract distribution for the same metric (n=30,595). Hohmann-suite and sensational-language metrics are computed at the abstract level only. The title falls below the readability scripts' minimum word count, so its classical scores are not reported. A percentile is not defined for a count of novel acronyms, which is shown as n/a. A = abstract, Pct = percentile.